Understanding Cost of Capital (WACC)

Understanding cost of capital

- **The cost of each source of finance to the company can be equated with the return which the providers of finance (investors) are demanding on their investment.**
- **The cost of equity (K_e or r_e) to the company is the return the investors (shareholders) expect to achieve on their shares.**
- **The *cost of debt* (K_d or r_d) is the interest on debt or the yield on debt instruments.**
- **The effective *cost of debt* is the after-tax cost**

WACC formula

For firm financed by equity and debt, the cost of capital is the weighted average cost of capital (WACC)

- $\text{WACC} = r_e\left(\frac{Ve}{Ve+Vd}\right) + r_d\,(1 - \text{tax})\left(\frac{Vd}{Ve+Vd}\right)$
- Ve/(Ve + Vd) = equity proportion
- Vd/(Ve + Vd) = debt proportion
- Equity proportion + debt proportion =1 (100%)
- *Note: Using the long-term sources of finance*

WACC formula

- $WACC = r_e\left(\frac{Ve}{Ve+Vd}\right) + r_d\,(1 - tax)\left(\frac{Vd}{Ve+Vd}\right)$
- r_e = cost of equity;
- r_d = cost of debt (gross)
- $r_d\,(1 - tax)$ = after-tax cost of debt
- Ve = value of equity;
- Vd= Value of debt

Activity 1

- **EK Co is currently 20% debt financed. The cost of equity is 15% while the cost of debt before tax is 10%.**
- **Company tax is 35%.** *Calculate WACC*
- ***Solution:***
- ***Debt proportion =20%,***
- ***Equity proportion = 80%***
- ***Cost of equity = 15%,***
- ***Cost of debt before tax = 10%, Tax = 35%***
- ***WACC = (15%*0.80) + 10%*(1 – 0.35)*(0.20)***
- ***= 13.3%***

Activity 2

$$WACC = r_e\left(\frac{Ve}{Ve+Vd}\right) + r_d\,(1 - tax)\left(\frac{Vd}{Ve+Vd}\right)$$

- **TC Co is currently financed by K60m equity and K20m debt. The cost of debt <u>after tax</u> is 8% while the cost of equity 12%.**
- **Company tax is 30%.** *Calculate WACC*
- ***<u>Solution:</u>***
- ***Value of equity = K60m,***
- ***Value of debt = K20m***
- ***Cost of equity = 12%,***
- ***Cost of debt after tax = 8%, Tax = 30%***
- ***WACC = 12%*(60m/80m) + 8%*(20m/80m)***
- ***= 11.0%***

Cost of individual capital elements

- **Equity shares = CAPM or DVM**
- **Preference shares = Yield on Preference shares.**
- **<u>Debt :</u>**
  - **Perpetual debt = Interest yield**
  - **Redeemable debt = IRR (YTM)**
  - [redacted]
  - **Bank loan = Interest attached to the loan**
  - **With debt beta = CAPM**

Cost of Equity (K_e or r_e)

- **Based on Capital Asset Pricing Model (CAPM);**
- $r_e = r_f + \beta*(r_m - r_f)$
- ***Where;*** r_f = **return on the risk-free asset (yield on government securities)**
- r_m = **return on the market portfolio**
- **β= beta coefficient or value**
- $r_m - r_f$ = **market risk premium**
- **Rf = 4%, rm = 10%, B = 1.20**
- **Re = 4% + 1.20*(10%-4%) = 11.2%**

Activity 3

- ***Calculate the cost of equity using CAPM given that;***
- ***The beta coefficient of a company's share is 0.85, market risk premium is 9% and return on the risk-free asset is 6.35%.***
- **re = 6.35% + 0.85*(9%) = <u>14.00%</u>**

Cost of Equity (K_e or r_e)

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- $r_e = r_f + \beta*(r_m - r_f)$
- ***Where;*** r_f = **return on the risk-free asset (yield on government securities)**
- r_m = **return on the market portfolio**
- β= **beta coefficient or value**
- $r_m - r_f$= **market risk premium**

Calculating the beta value of an asset (2)

- $$\text{Beta x} = \frac{Correlation\ x\ with\ market * Std\ dev\ of\ return\ x}{std\ dev\ of\ market\ return}$$
- $$= \rho * \frac{\sigma_x}{\sigma_m}$$
- ρ is the correlation coefficient between asset x and the market.
- σ_x is the standard deviation of asset x
- σ_m is the standard deviation of the market returns (m)

Activity 4

- **Calculate the beta of FJ plc, which is listed on LuSE. Based on data over the past ten years, the correlation between FJ plc, and LuSE is 0.90. FJ plc has a standard deviation of returns of 62.80% and LuSE has a standard deviation of returns of 53.41%.**
- **The yield on treasury bills is 6%, average return on LuSE is 13.0%. FJ pays tax at 25% while the firms corporate debt gross yield is 8%. FJ is currently 20% geared.**
- **<u>Required:</u>**
- **Calculate the cost of capital for FJ plc**

Extracts from activity 4

Variable	Value (figure)
Correlation coefficient	0.90
Std deviation of the FJ returns	62.80
Std deviation of LuSE returns	53.41
Risk free rate of returns	6.0%
Average returns on LuSE	13.0%
Gross yield on debt	8.0%
Corporate tax rate	25.0%
Gearing ratio (debt ratio)	20.0%

Solution to activity 4.

- Beta Fj = $\frac{0.90 * 62.80}{53.41}$ = **<u>1.058</u>**
- Cost of equity = rf + B*(rm – rf)
- = 6% + 1.058*(13%-6%)
- = **<u>13.41%</u>**
- WACC = 13.41%(0.80) + 8%*(1 – 0.25)(0.20)
- = 10.73% + 1.20%
- = **<u>11.93%</u>**

Cost of Equity (K_e or r_e) cont

- **Based on *Dividend* Valuation Model (DVM);**
- **The cost of equity is the *discount rate* which equates the sum of the present values of all future dividends to the current share price.**
- ***Assumptions on dividend are;***
  - **Dividend is constant i.e. forms a perpetuity**
  - **Dividend growth rate is constant (g)**
  - **Non-constant growth**

Cost of equity (r_e) based on constant dividend

- $$r_e = \frac{D}{Po\ (ex-div)} \times 100$$

- **Where;**
- r_e = cost of equity or return on equity
- P_o =current share price (ex-dividend) = *Market price*
- D = Constant dividend
- P_o (ex-div) = P_o(cum-div) - D

Cost of equity (r_e) based on constant dividend growth.

- $$r_e = \frac{d_0\,(1+g)}{P_0\,(ex-div)} + g \quad \text{or} \quad = \frac{d_1}{P_0\,(ex-div)} + g$$

- **Where;**
- d_o = dividend paid or payable now (t = 0)
- g = Constant growth rate
- d_1 =dividend payable at the end of the first period
- Po(ex-div) = share price that excludes current dividend.

r_e; based on non-constant dividend growth.

- **Based on the historic growth rate;**

$$g = \sqrt[n]{\left\{\frac{Latest\ dividend}{Earliest\ dividend}\right\}} - 1$$

- **g = approximate growth rate.**
- **n = period of growth or intervals between dividends.**
- **Cost of equity is then calculated using the constant growth rate assumption with d_o**

r_e; based on non-constant dividend growth.

- Based on the historic growth rate;

$$g = \sqrt[n]{\left\{\frac{Latest\ dividend}{Earliest\ dividend}\right\}} - 1$$

- g = approximate growth rate.
- n = period of growth or intervals between dividends.
- **Task 1:**
- Determine the average growth rate given the following:
- Year 1 2
- Dividend 12n 15n **Answer = 25%**
- **Task 2:**
- Determine the average growth rate given the following:
- Year 1 2 3 4
- Dividend 12n 15n 18n 19n **Answer = 16.55%**

Activity 5

- **A business is financed by K60m equity and K40m debt. The cost of debt before tax is 10% while the dividend just paid is 4.2 ngwee per share. The dividend has grown steadily from 2.0 ngwee paid 5 years ago. The current share price is 97.5 ngwee per share. Company tax is 30%.**
- ***<u>Required:</u>***
- ***Calculate the cost of equity and determine the WACC.***

Solution to activity 5.

- $$g = \sqrt[5]{\left\{\frac{4.20}{2.0}\right\}} - 1 \quad = 16\%$$
- $$re = = \frac{4.20*(1.16)}{97.5} + 0.16 = 21.0\%$$

Capital element	Cost (%)	Proportion	Cost * Proportion
Equity	21	60m/100m	12.60%
Debt	10*(1-0.3)	40m/100m	2.80%
		WACC	15.40%

r_p; Cost of Preference shares.

- Preference shares are issued with a fixed rate of dividend. Assuming the shares are irredeemable, dividend forms a perpetuity. Thus;

- $r_p = \frac{D}{Po\ (ex-div)}$ x 100

- Where;

- r_p = Yield on preference shares or cost of preference shares

- P_o =current preference share price (ex-dividend)

- D = dividend on preference shares

r_b; Cost of a bank loan.

- Bank loans are non tradable. They are issued with a specific rate of interest fixed or variable.
- Since interest is tax deductible, the effective cost of a bank loan is the after tax cost calculated as;
- After tax cost = pretax cost (1 – tax rate)

r_d; Cost of irredeemable bonds.

- Irredeemable bonds are issued with a fixed rate of interest and no maturity date.
- Interest payable forms a perpetuity.
- $r_d = \frac{C*(1 - tax\ rate)}{Po\ (ex-int.)}$ x 100
- C = (coupon payment) = Coupon rate x par value
- Po = market price of the bond (ex-int.)

r_d; Cost of redeemable bonds (YTM).

- Redeemable bonds are issued with a fixed rate of interest and a specific maturity date.
- The cost of this instrument is the yield to maturity (YTM) or internal rate of return(IRR).
- Use of interest net of tax produces an after tax cost of debt.

r_d; (YTM) formula.

- $$\text{YTM (IRR)} = A + \left\{\frac{a}{a-b} * (B - A)\right\}$$
- **<u>Where;</u>**
- **A = lower discount rate with positive NPV (a)**
- **B = Higher discount rate with negative NPV (b)**
- **A = 5%, a = K2.75; B = 7%, b = -K2.74**
- **IRR = 5% + {2.75/(2.75 –(-2.74)*(7%-5%)}**
- **=5% + (2.75/5.49*(2%)) = <u>6.00%</u>**

r_d; Cost of debt based on CAPM.

- **Where debt beta is known, the *cost of debt* <u>*before*</u> *tax* can be calculated using CAPM as;**
- $rd = rf + \beta d*(rm - rf)$
  - ***Where*: βd =debt beta**
- **After tax cost = pretax cost (1 – tax rate)**

Activity 6

Miguel Business Solutions (MBS) is funded with 80m (K1) ordinary shares, 8% K30m preference share capital and K40m of debt trading at par. Preference shares are irredeemable and quoted at K85 per K100 nominal value. The dividend growth rate at the company is constant at 6% per annum forever. Latest dividend is 20 ngwee per share and the share price is 212n. The debt beta is 0.75. The company faces a tax rate of 30%. The risk-free interest is 4.6% and the stock market risk premium is 6%.

<u>Required:</u>

Calculate WACC for MBS.

Solution to activity 6 (1):

- Cost of equity = $\frac{20*(1.06)}{212} + 0.06 = 16\%$
- Cost of preference share = $\frac{8\%*K100}{85} * 100 = 9.41\%$
- Cost of debt (CAPM) = 4.6% + 0.75*(6%) = 9.10%

Value of capital elements:	K'm
Equity = 80m * K2.12	= 169.60
Debt (trading at par)	= 40.00
Preference shares (K30m/K100)*K85	= 25.50
Total capital	235.10

Solution to activity 6 (2):

Capital element	Value K'm	Proportion	Cost %	Cost * proportion
Equity	169.6	0.7214	16.00	11.54
Preference	25.50	0.1085	9.41	1.02
Debt	40.00	0.1701	9.1*0.7	1.08
	235.10		WACC	13.64%

Solution to activity 6 (1):

- **Alternatively:**

- $$\text{WACC} = \frac{(16\% * 169.6m) + (9.41\% * 25.5m) + (9.10\% * 0.7 * 40m)}{235.10}$$

- $$= \underline{13.64\%}$$